AP CALCULUS AB

Unit 1 • Self Check 1.1 and 1.2 • Lessons 1-1 to 1-4

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Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$$\lim_{x \to -3} \frac{x^2 + x - 6}{x^2 - 9} =$$

(A) $\frac{5}{6}$

(B) 0

(C) $-\frac{5}{6}$

(D) It does not exist.

Question 2

NO CALCULATOR

$$\lim_{x \to \infty} \frac{2x - x^3}{4x^3 + 1} =$$

(A) 2

(B) $-\frac{1}{4}$

(C) 0

(D) It does not exist.

Question 3

NO CALCULATOR

Let $j(x) = \begin{cases} x + 3, & x < 1 \\ 2x, & x \geq 1. \end{cases}$ Then $\lim_{x \to 1} j(x) =$

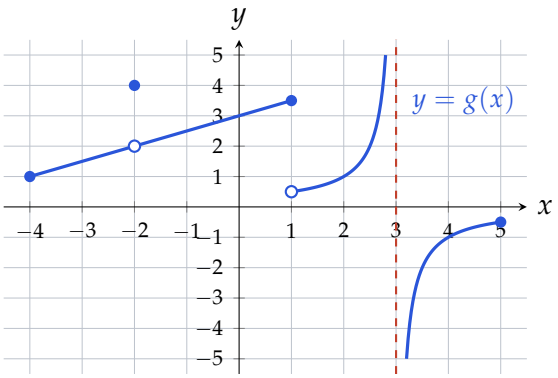
(A) 2

(B) 4

(C) 3

(D) It does not exist.

Question 4 and the free response refer to the graph of g.



Question 4

NO CALCULATOR

Using the graph of g, at which x is the discontinuity removable?

(A) $x = -2$
(B) $x = 1$
(C) $x = 3$
(D) $x = -4$

Question 5 (Free response)

NO CALCULATOR

Using the graph of g, classify the discontinuities at $x = -2$, 1 and 3, with limit statements.